# GLIMPSE INTO ONLINE EDUCATION ON coursera

STATS401 Data Acquisition and Visualization
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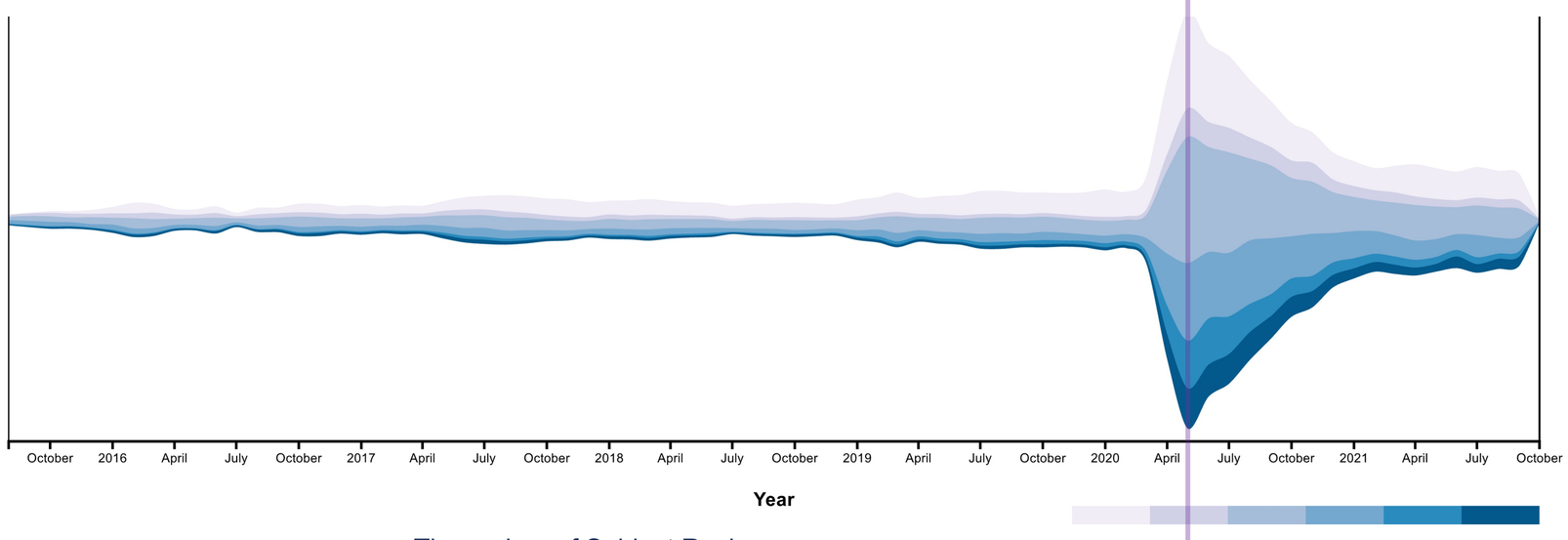
## April, 2020

**Covid-19 Global Outbreak**

## INTRODUCTION

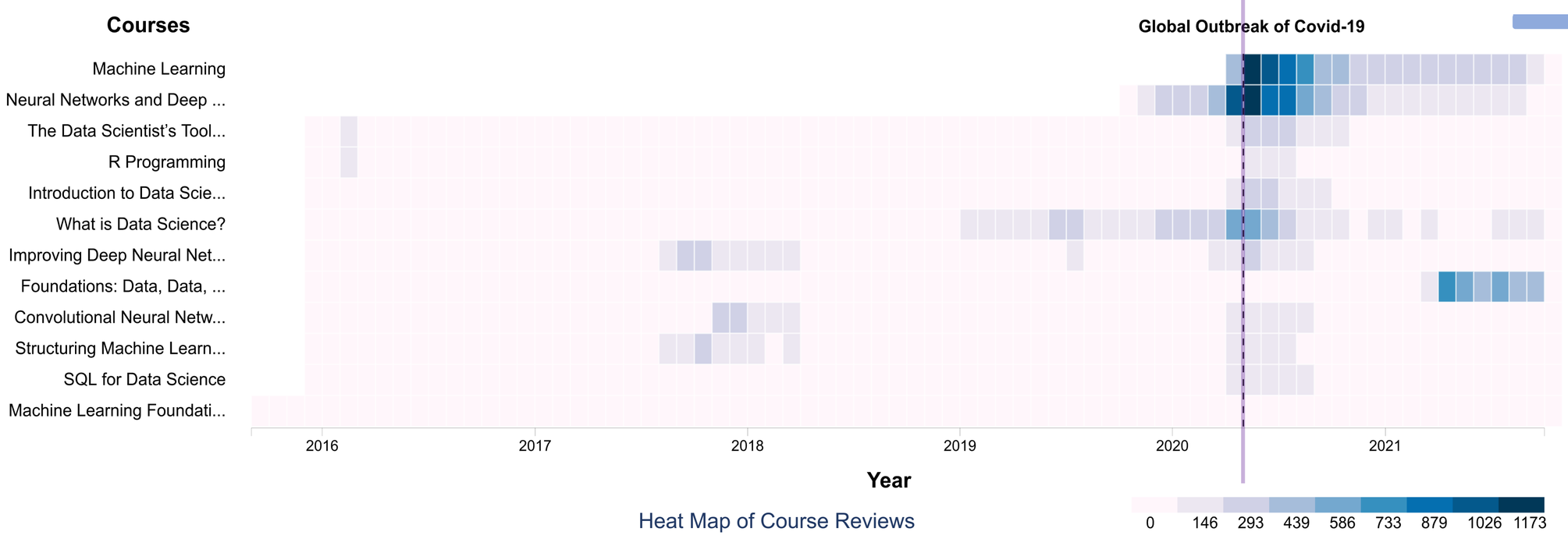
**Online education** has the advantage of allowing students to participate in high-quality courses when distance and schedule make on-ground learning difficult. This advantage became obvious since the outbreak of the global pandenmic Covid-19. **Coursera** is one of the biggest online course provider. It collaborates with more than 200 universities and companies and has more than 77 million users worldwide.
This project researches into the **situation of online education on Coursera, focusing on Data Science department.**




Themeriver of Subject Reviews

## THEMERIVER

How is the popularity of courses on **Coursera** by examining the monthly increasement of reviews in 6 fields?
After the outbreak of Covid-19 in Feb 2020, there is a significant increasement in the review number in all 6 fields, especially Data Science and Business which have involved large amounts of courses for years. Particularly, many Health courses focused on Covid-19 open. The number of increasement of reviews peaks in May 2020 and begin to stabilize in Jan 2021. It is hypothesized that staying at home prompts many people to pay attention to online education.




Heat Map of Course Reviews

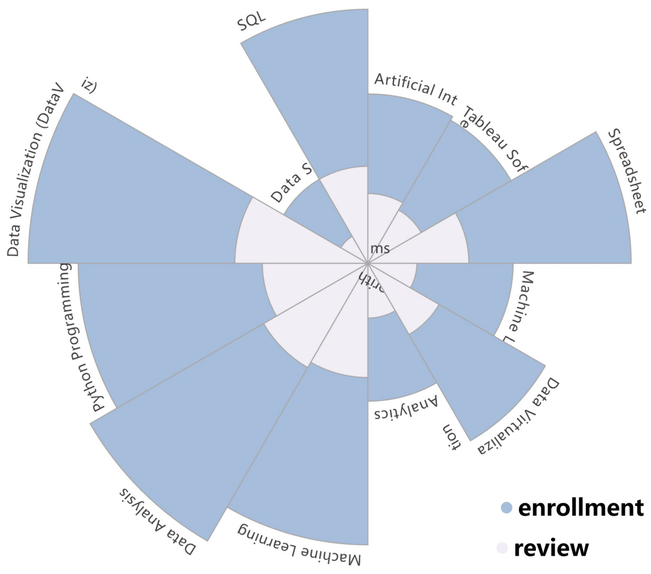
## HEAT MAP

How does the **popularity** of **courses** of different subjects change through time?
The most popular courses in these six subjects all have a **boost** in their reviews after the global outbreak of Covid-19. The course "COVID-19 Contact Tracing" opened and "The Science of Well-Being" thrived soon after the outbreak. Meanwhile, together with plenty of language learning courses, many practical introductory courses of Data Science and Business gained attention.

## ROSE CHART

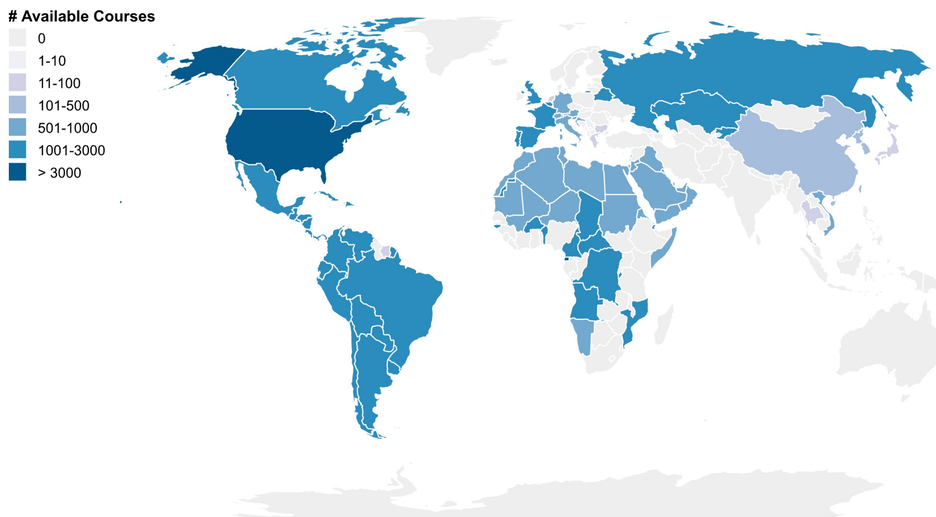
What are the most **popular tags** in Data Science on **Coursera** by investigating the sum of enrollment numbers each hot tag involves?

Among the most popular 12 tags in Data Science, Data Visualization (DataViz), Data Analysis, and Python Programming are of the greatest interest, each of which has attracted over 1 million users so far. It is speculated that **Data Visualization** receives great attention because of its extensive and flexible adaptation in different fields. Moreover, machine learning with over 0.9 million students follows, which is in line with the boom of AI. The review number is generally proportional to the enrollment number as well.




Rose Chart of Popular Tags

## MAP




Map of Course Resources

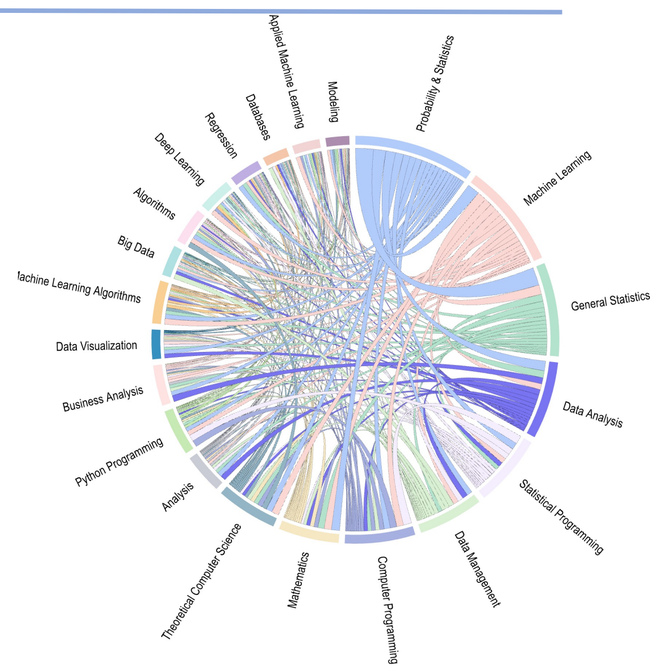
How do online education **resources distribute** around the world?
With its primary language (**English**) and one of its main languages (**Spanish**) being the most popular languages on **Coursera** , the US has the most course resources. Users from Russia, Canada, and South America also have plentiful subtitles in their favor.

## CHORD DIAGRAM

What pairs of **tags co-occur** most frequently in Data Science department?

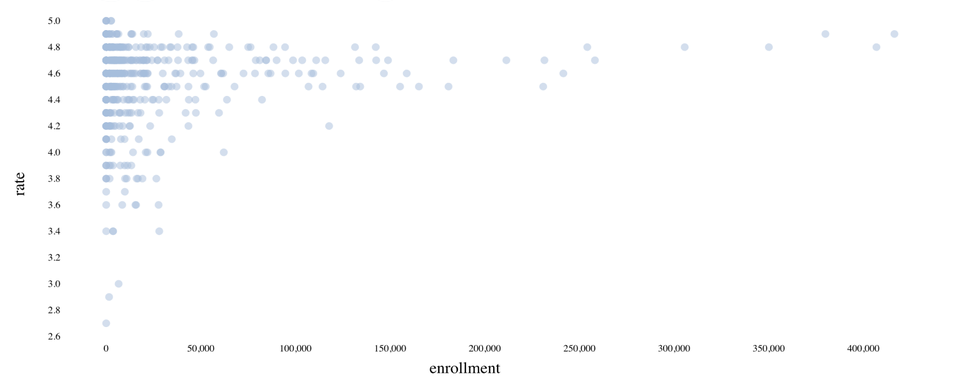
Pairs of tags that co-occur frequently:
(Probability, Analysis);
(Machine Learning, Computer Programming);
(Algorithm, Computer Science);
(Data Visualization, Data Analysis)…

The above-mentioned pairs are jointed offered in large number of courses, indicating that the combined skills are very useful and should be learned together. For example, to be good in machine learning, one also needs to learn about computer programming; to be good at data visualization, one would definitely need the skills of data analysis…




Chord Diagram of Co-occuring Tags

## SCATTER PLOT




Scatter Plot of Enrollment Number & Rating Score

How are courses distributed with regards to basic features like rating, difficulty level, number of enrollment, number of reviews, required length?
This graph indicates that the courses with the **highest** number of enrollment have **high** rating score of above 4.7 while the courses with rating score under 3.0 tend to have very few students enrolled.

## CONCLUSION

After Covid-19 global outbreak:

- **More users** have been drawn to online education platforms due to health concerns and the limited access to in-person teaching;
- **Introductory level courses** become increasingly popular, probably because people have a lot of free time at home and want to learn some skills;
- **Public health courses** receive much more attention than before, the course: Covid-19 Contact Tracing quickly became one of the most popular courses in **Coursera**;

Among courses offered by **Coursera**:

- The most **popular tags** in Data Science include Machine Learning, Data Visualization, Data Analysis and Python Programming;
- Some **combined skills** are very important, like Data Visualization and Data Analysis, Machine Learning and Computer Programming…
- **English-speaking** and **Spanish-speaking** countries have relatively more online academic resources compared with other countries, followed by Russian-speaking and Portuguese-speaking areas.